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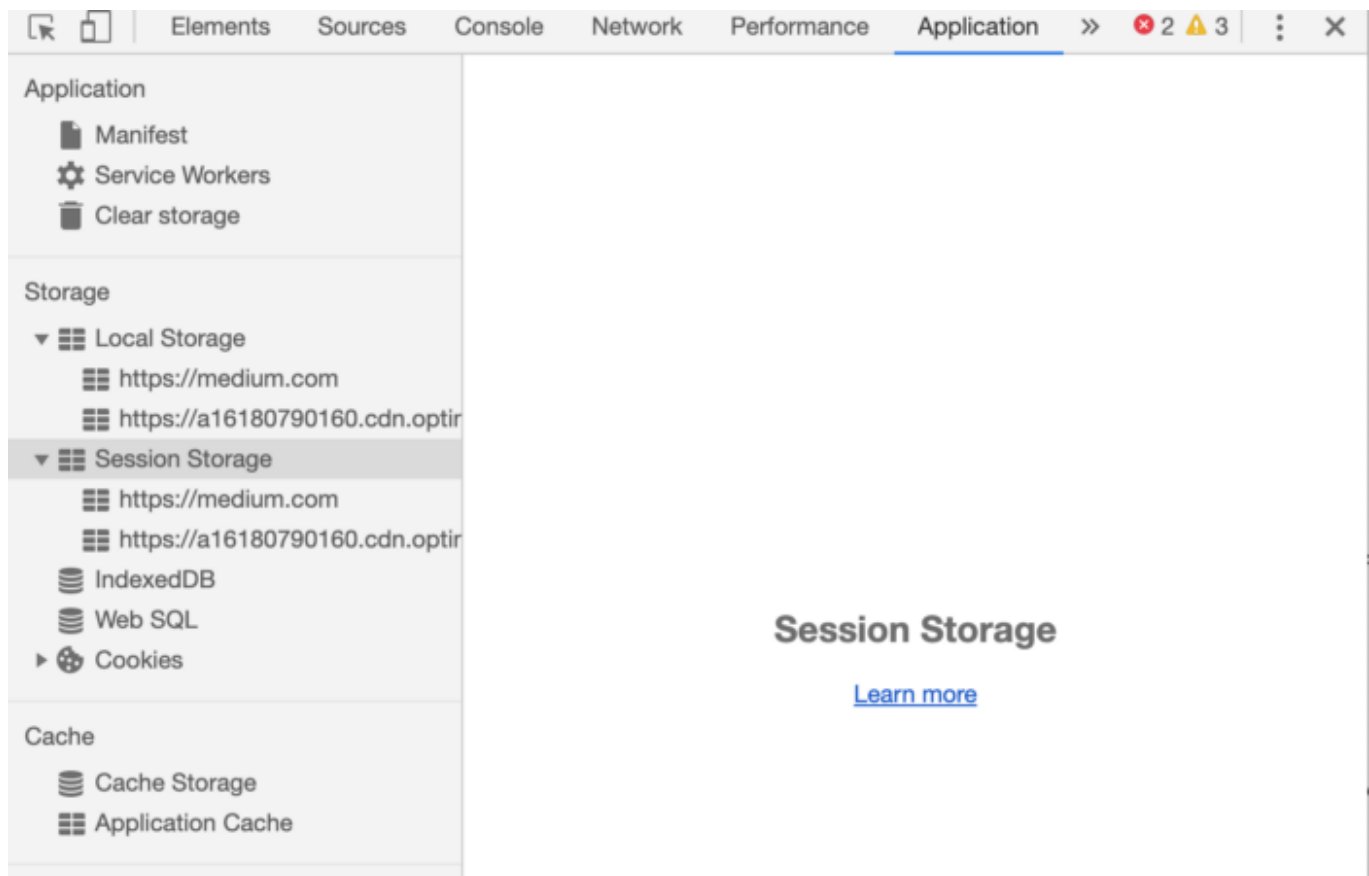
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Local Storage vs Session Storage vs Cookie



krishankant singhal May 19, 2020 · 3 min read ★



Various Storage provided by browser now

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cookies was very small. but now we local storage and session storage as well. and cookies has been talk of past , though it is getting used for various purposes. let's talk about all these

LocalStorage

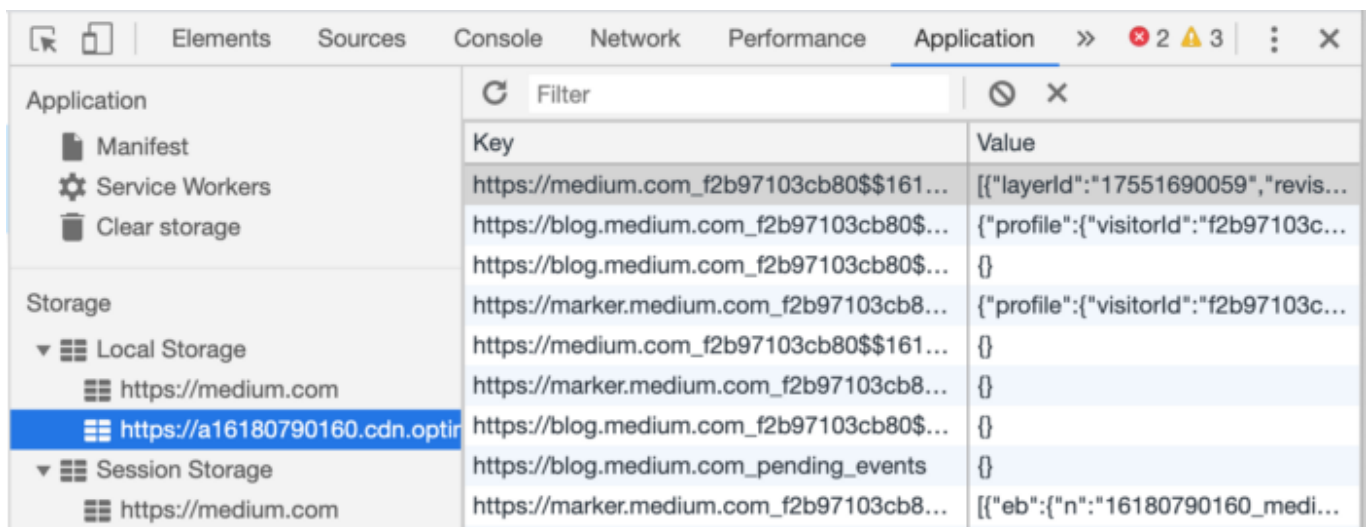
localStorage is a way to store data on the client's computer. It allows the saving of key/value pairs in a web browser and it stores data with no expiration date. localStorage can only be accessed via JavaScript, and HTML5. However, the user has the ability to clear the browser data/cache to erase all localStorage data.

Pros:

- stores data with no expiration date
- storage limit is about 5MB
- data is never transferred to the server

Cons:

- plaintext, hence not secure by design
- limited to string data, hence need to be serialized
- can only be read on client-side



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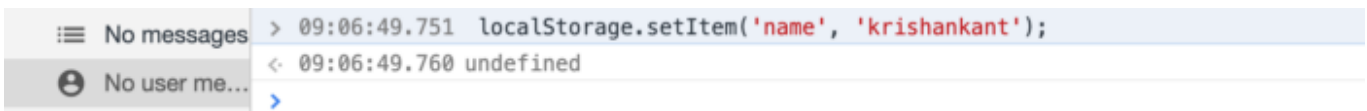
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Local Storage

So we see , we have key value storage, where we have key and value as an serialized object.

We can try localStorage api from console



and it will create a key value pair in localStorage store.



Session storage

- stores data only for a session, meaning that the data is stored until the browser (or tab) is closed
- data is never transferred to the server
- can only be read on client-side
- storage limit is about 5-10MB

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Key	Value
branch_session	{"session_id": "79110609450356..."}
lightstep/clock_state/collector-medium.ligh...	{"timestamp_micros": 15898191..."}

```
{
  "session_id": "791106094503563413",
  "identity_id": "7911060945206019...",
  "browser_fingerprint_id": "NDYyMjU3MjY2OTA4OTAwMzIS",
  "data": "{\"+clicked_branch_link\":false,\"+is_first_session\":true}",
  "has_app": false,
  "identity": null,
  "identity_id": "791106094520601988",
  "link": "https://link.medium.com/a/key_live_ofxXr2qTrrU9NqURK8ZwEI",
  "session_id": "791106094503563413"
}
```

Same as localstorage , we can write key-value pair in session storage as well.

```
> 09:11:21.139 sessionStorage.setItem('lastname', 'singhal');
< 09:11:21.146 undefined

>
```

Key	Value
lastname	singhal
branch_session	{"session_id": "791106094503563413", "identity_id": ...}

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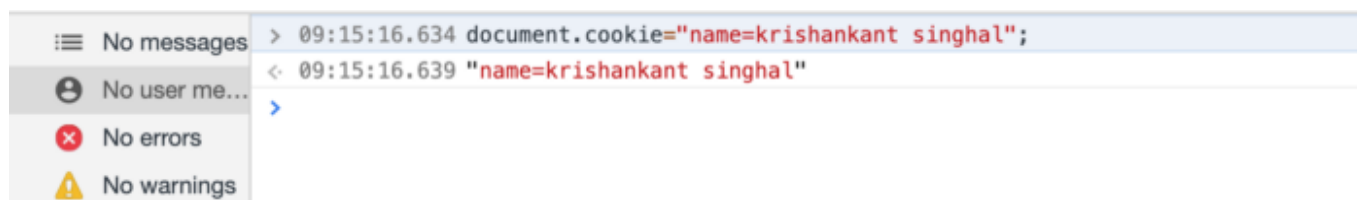


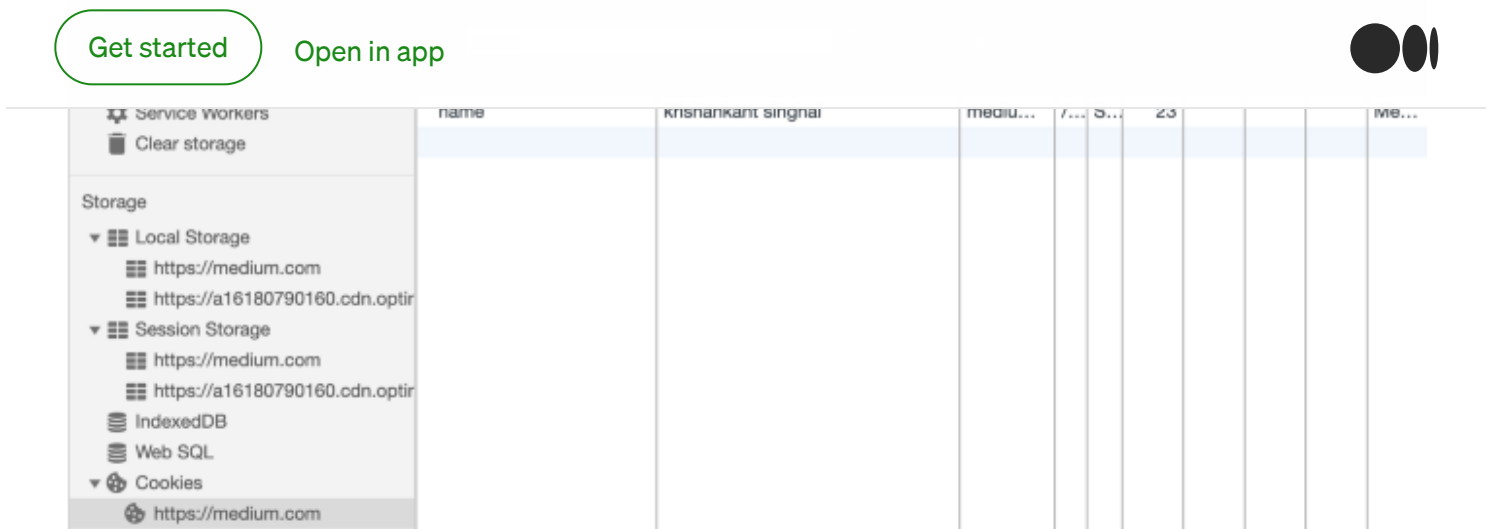
Cookie

- Stores data that has to be sent back to the server with subsequent XHR requests. Its expiration varies based on the type and the expiration duration can be set from either server-side or client-side .
- Cookies are primarily for server-side reading (can also be read on client-side), localStorage and sessionStorage can only be read on client-side.
- Size must be less than 4KB.
- Cookies can be made secure by setting the httpOnly flag as true for that cookie. This prevents client-side access to that cookie.

Name	Value	Domain	P.	E..	Size	Http...	Se...	Sa...	Pri...
sz	1260	mediu...	/	2...	6				Me...
lightstep_guid/lite-web	75daa36967c3c581	mediu...	/	2...	39				Me...
g_state	{ "l": 0 }	mediu...	/	2...	16				Me...
tz	240	mediu...	/	2...	5				Me...
lightstep_guid/medium...	dd33ba419cc816a0	mediu...	/	2...	41				Me...
_parsely_session	{%22sid%22:722%2C%22surl...	.mediu...	/	2...	222				Me...
_parsely_visitor	{%22id%22:%22pid=b81d5127...	.mediu...	/	2...	133				Me...
_gid	GA1.2.1903342222.1589470086	.mediu...	/	2...	31				Me...
pr	2	mediu...	/	2...	3				Me...
__cfuid	9299db59a4e06f2b405ef37cbeff...	.mediu...	/	S...	59	✓	✓	None	Me...
lightstep_session_id	973d491ecfe9cf1	mediu...	/	2...	35				Me...
_s	owHsiKtEtEdnRy08W%2F4yEEa...	.app.link	/	2...	68			✓	None
sid	1:mDC09Fu8zY8PVj0Ovb/HmAj...	.mediu...	/	2...	69	✓	✓	None	Me...
optimizelyEndUserId	oeu1464639966643r0.91045748...	.optimi...	/	2...	54				Me...
uid	f2b97103cb80	.mediu...	/	2...	15	✓	✓	None	Me...
nid	6053a4c8f54dc35ba75468af45d...	.nival...	/	2...	35				Me...

Cookies can be updated , set using document.cookie object from browser window object.





In chrome browser, by inspect element, we can go to application tab and see values which are in localStorage, Session Storage and Cookies.

So in nutshell

Feature	Cookies	localStorage	sessionStorage
Maximum data size – more info	4 kB	5 MB	5 MB
Blockable by users – more info	yes	yes	yes
Auto-expire option – more info	yes	no	yes
Supported data types – more info	string only	string only	string only
Browser support – more info	very high	very high	very high
Accessible server-side – more info	yes	no	no
Data transferred on every HTTP request – more info	yes	no	no
Editable by users – more info	yes	yes	yes
Supported on SSL – more info	yes	n/a	n/a
Can be accessed on – more info	server-side &	client-side	client-side only

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	automatic		
Lifetime – more info	as specified	till deleted	till tab is closed
Secure data storage – more info	no	no	no

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